

Know Your IC: 555 Timers

by [frenzy](#) on June 11, 2010

Table of Contents

Know Your IC: 555 Timers	1
Intro: Know Your IC: 555 Timers	2
Step 1: History Of The Chip	2
Step 2: Function of the Chip	3
Step 3: Project: 555 timer Light Blinker	4
Step 4: Project: Layout and Schematic	4
Step 5: Project: Analysing the circuit	5
Step 6: Final Thoughts	6
Related Instructables	6
Comments	6



Author: **frenzy** Groovy Pancakes

I'm an Intern here at Instructables. I help keep the site running smoothly and send you t-shirts! I also build cool things. Follow me
<http://twitter.com/greggawatt>

Intro: Know Your IC: 555 Timers

Hello and welcome to a new instructables series called "Know Your Integrated Circuit"

Know Your IC seeks to demystify common Integrated Circuits and allows people to understand them to a point where they can use them in their own projects.

The first circuit in this series is the widely used 555 timer.

If you have suggestions for future chips we should cover, please shoot me an email!

UPDATE:

I decided to make the layout pictures easier to read, i know it was confusing some people. I hope this helps!



Step 1: History Of The Chip

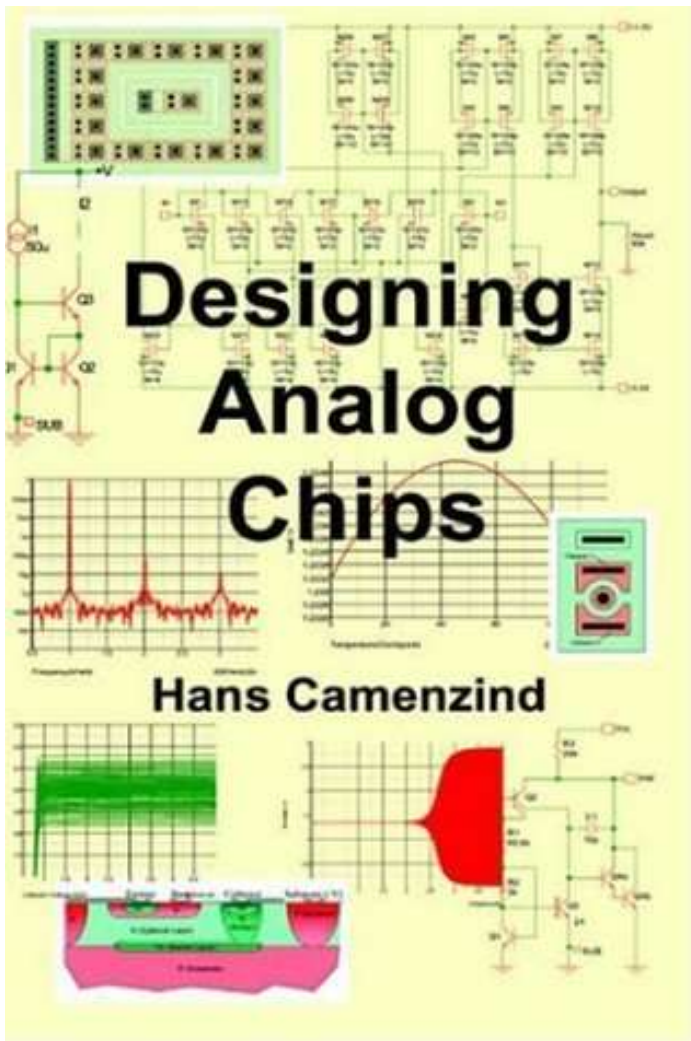
The 555 timer was created by Hans Camenzind (b. 1934) in 1970 while he was working for Signetics. The design process took about a year. Designing the chip was all done by hand, using rubylith which made the process take much longer than by today's standards.

The 555 timer replaced 23 transistors, 15 resistors and 2 diodes and allowed manufacturers to use this problem solving method much easier.

Today with billions of the chips in existence, it is one of the most widely used chips. Most people who do electronics hear of the 555 timer first. It is the gateway drug to the world of integrated circuits.

References

<http://www.semiconductormuseum.com/Transistors/LectureHall/Camenzind/>



Step 2: Function of the Chip

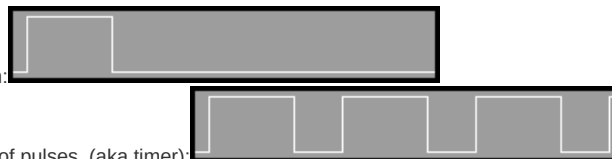
The 555 integrated circuit is commonly called the "555 timer" but that is only one of its uses. The chip can be used both as a timer and as an oscillator.

WHAT DOES THAT MEAN?

Well i'm glad you asked. An Oscillator is something that produces a constant electrical wave, like a sine or square wave. The timer will pulse at a specific interval depending on the components attached.

The 555 chip has 3 main modes:

- Monostable Mode for "One Shot" pulse generation:
- Astable Mode for putting out a continuous stream of pulses. (aka timer):
- Bistable mode for acting like a Schmitt trigger.

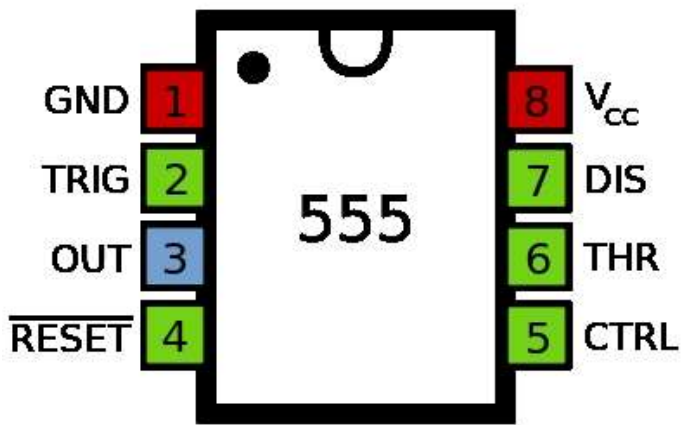


The Pin out of the 555 timer is as follows:

- Pin 1: Ground
- Pin 2: Triggering the timer, connecting this to ground starts the chip up.
- Pin 3: Output, stays at whatever Vcc is
- Pin 4: Forces pin 3 to low if grounded
- Pin 5: Used to adjust trigger threshold.
- Pin 6: Threshold that ends timer when 2/3 of Vcc is reached.
- Pin 7: Discharge, connects to ground when output is low
- Pin 8: Power, denoted in most contexts as Vcc

Images above are borrowed from: <http://clarkson-uk.com/555-timer/>

Resources: <http://www.markallen.com/teaching/ucsd/147a/lectures/lecture4/5.php>



Step 3: Project: 555 timer Light Blinker

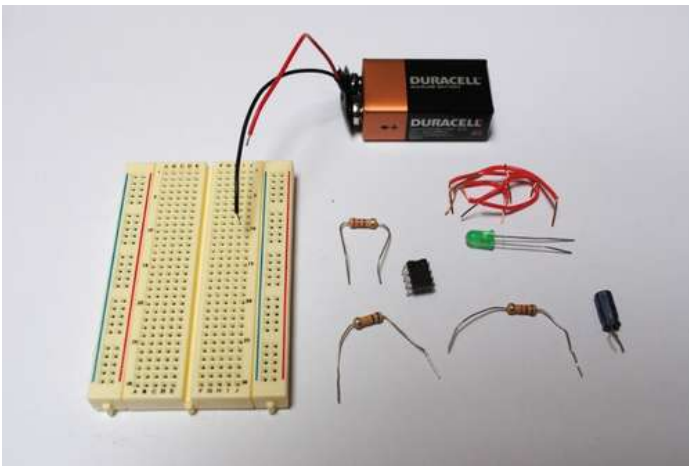
For our project i will be building a simple LED flasher.

For this project you will need the following:

- 1 555 chip (Radio shack sells them, or get them for 25 cents each online)
- 1 100K Ohm Resistor
- 1 12k Ohm Resistor
- 1 220 Ohm Resistor
- 1 4.7 uF Capacitor
- 1 LED
- 1 9v Battery Snap
- 1 9v Battery

Then to lay it out you will want:

- 1 Breadboard
- some solid core wire for hook ups.

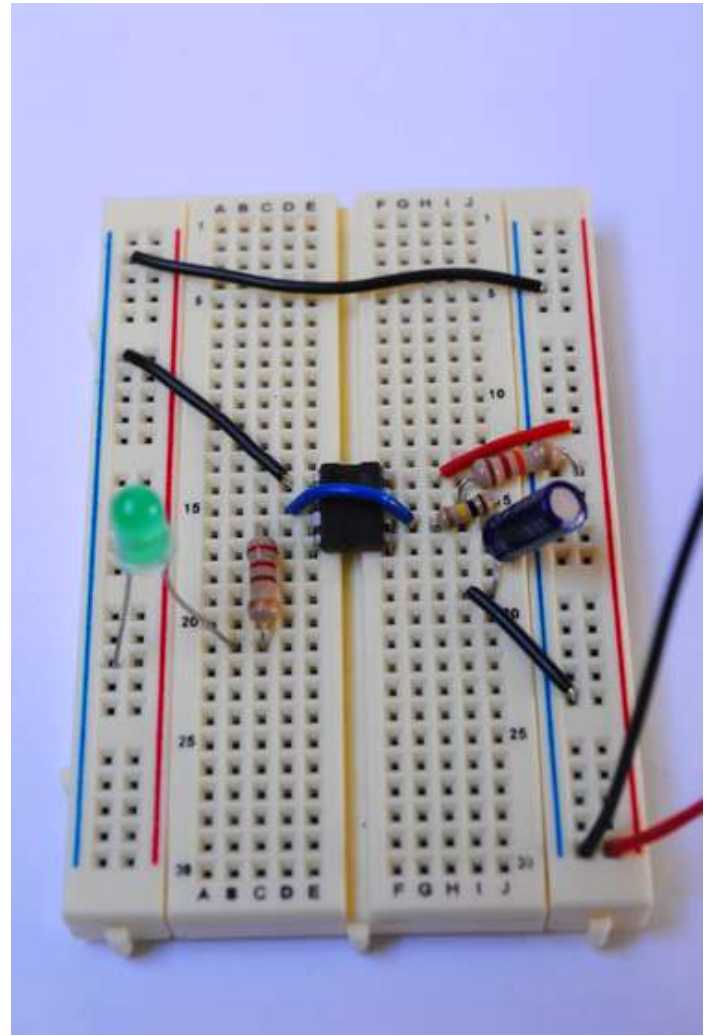
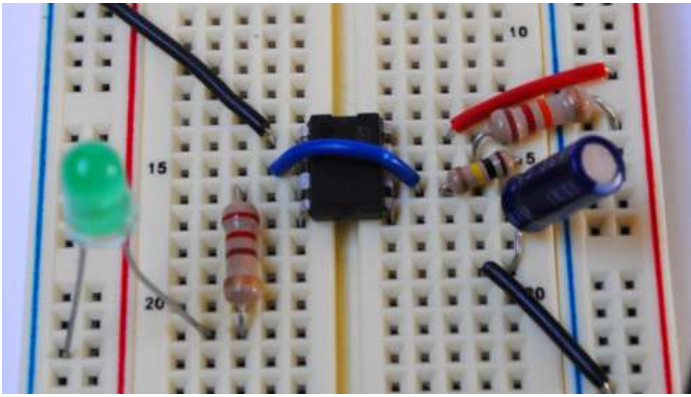


Step 4: Project: Layout and Schematic

We will go over the layout by pin number, Just do the following the the circuit should work, if it doesn't take a look at my breadboarded version

- Pin 1: Ground
- Pin 2: goes to pin 6
- Pin 3: hooks up to the 220 ohm resistor, then to the led and then the ground of the led goes to ground.
- Pin 4: Unused
- Pin 5: Unused
- Pin 6: Has the 100k Ohm resistor between it and pin 7 and the 4.7 uF capacitor from it to ground (Make sure you get the polarity right!).
- Pin 7: Has the 100k Ohm resistor between it and pin 6 and a 12k resistor that goes to power
- Pin 8: Power

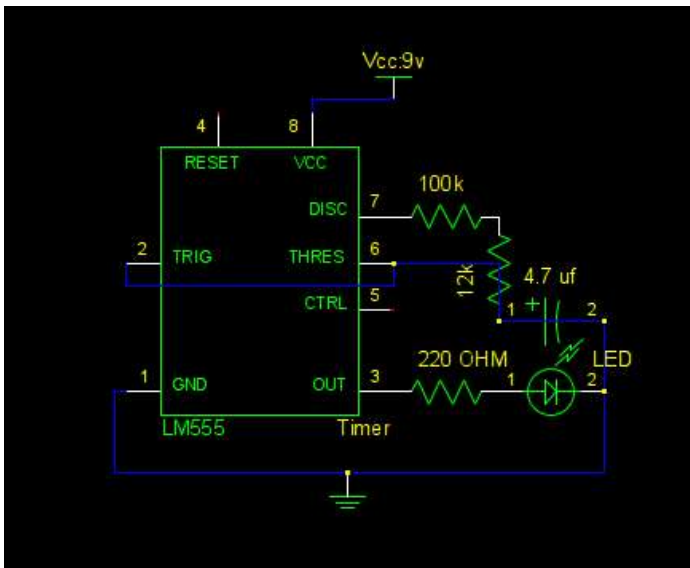
Once that is put together, test it out. The LED should blink on and off slightly quick. again if something doesn't seem right check the pictures and if your IC gets hot remove power immediately or you'll fry your chip.



Step 5: Project: Analysing the circuit

If you understand what the pinouts of the 555 timer chip mean, you can see what is happening with this circuit. This project is a simple implementation of this chip.

The capacitor controls the blink speed so you could change that out to make it blink slower or faster. Higher capacitance = lower speed.



Step 6: Final Thoughts

I hope you enjoyed this first edition of know your IC. If you have suggestions for this series or would like to do you own "Know Your IC" installment, Please contact me!



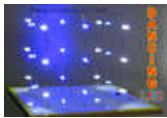
Related Instructables



learn about the 555 by Jimmy Proton



41 LED Flasher Circuit using 555 IC by supes



DANCING LED's by Dipankar



Time Lapse Intervalometer for SLRs with 555 timer IC by eagleapex



"Dark Detector" using a 555 timer IC by dandeeman



Mobile Phone Alarm for a motorbike, car or anything you want by simon72post

Comments

35 comments

[Add Comment](#)



Jeffery Mouse says:

I like this series! This is a great help to a novice like me!

Feb 28, 2011. 1:55 PM [REPLY](#)



shubhneesh says:

bravo... i like it very much.

Feb 13, 2011. 8:39 AM [REPLY](#)



spikedeath69 says:

At radio shack there were two kinds of 555 timers and im not sure which to get also is the resistance of the resistor 1/4 watt or what? please reply thank you.

Feb 1, 2011. 7:45 AM [REPLY](#)



spikedeath69 says:

what are the volts on this capacitor?

Feb 1, 2011. 8:06 AM [REPLY](#)



protostack says:

Great 555 astable calculator over here http://www.ohmslawcalculator.com/555_astable.php

Jan 13, 2011. 1:21 PM [REPLY](#)



ziga321 says:

have been searching for LED blinking all day and this is the first one that actually DOES work. Thanks-a-lot!

Dec 26, 2010. 4:19 PM [REPLY](#)



wareneutron says:
that's i know

Nov 19, 2010. 1:40 AM [REPLY](#)



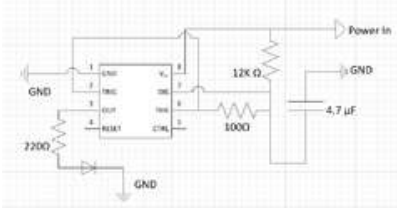
lordxenosi says:
I would love to see more of these 'Know Your IC' instructables. I have little experience with common ICs and I found this simple tutorial to be educational and enjoyable.

Nov 3, 2010. 12:45 AM [REPLY](#)



Genderle says:
Here is a better schematic

Oct 28, 2010. 7:49 AM [REPLY](#)



frikkie says:
I personally love the 555 chip there so many circuits one can build with it.
I build my first light effect with the same circuit as the author used.

Oct 22, 2010. 4:13 AM [REPLY](#)



YOGESH YADAV says:
what are the adjustable timer circuit of 555.....???

Oct 19, 2010. 5:47 AM [REPLY](#)



vince29 says:
i agree with xtank5, SW, and rabbitkillrun. there are a few incionsistancies between the text and the photos. When I first put it together the LED stayed on w/o blinking. In trying to correct this I m nable to get the LED lit again.
This could be the start of a very interesting development but answers should be forthcoming.

Oct 11, 2010. 11:19 AM [REPLY](#)



waldosan says:
i really like this series, it's opening my mind up to different ways you can use electronics.

Oct 7, 2010. 10:16 PM [REPLY](#)



Abume4life says:
please i need a step-by-step guide on lm741 on bread board

Sep 29, 2010. 2:53 AM [REPLY](#)



suckafish says:
use google.....

Oct 7, 2010. 11:46 AM [REPLY](#)



Abume4life says:
i have lm741 cant arrange the other component cause i dont have the circuit diagram
heeeeeeeeeeeeeeeelp

Sep 29, 2010. 3:13 AM [REPLY](#)



xtank5 says:
I can't get it to work. All of my connections are good, they are just as you describe, but my LED just stays on. I've tried using larger caps to slow it down in case it was blinking faster than I could see, but even with 100uf it just stays on steady. Are you sure your plans work? Help would be nice.

Sep 12, 2010. 2:58 PM [REPLY](#)



frenzy says:
Hello,

Sep 13, 2010. 4:49 PM [REPLY](#)

I updated the layout picture so it's easier to read, and i tested it out and it works. Post some good pictures of your layout and i can try to help you!



arhodes18 says:
I like this i'ble, but i wish there was just a little more explanation as to what the pins do. and yes i agree that the next should be LM386

Sep 9, 2010. 5:02 PM [REPLY](#)



Dream Dragon says:
Yes, I'd like to vote for an LM386 guide too.

Sep 13, 2010. 4:20 PM [REPLY](#)



rabbitkillrun says:
I agree with this.

Sep 10, 2010. 7:51 AM [REPLY](#)



8v92 says:
Good project! If you need equal on/off times a 74xx or 40xx counter can be used with the LM555 as long as Vcc is at 5 volts.

Sep 12, 2010. 6:53 PM [REPLY](#)



phenoptix says:
Nice guide. It seems the consensus is the LM386 for the next IC. I've had a lot of requests regarding ZE002 IC chips or what seem to be the same in the ANA608 that are used in garden solar lights to give a Joule Thief style circuit. Not been able to source them and don't have the knowledge of ICs to work out what they are.

Sep 12, 2010. 10:54 AM [REPLY](#)



wareneutron says:
i dont understand you work..

Sep 12, 2010. 2:32 AM [REPLY](#)



wareneutron says:
what is this

Sep 12, 2010. 2:31 AM [REPLY](#)



kevinhannan says:
I like this - a lot.
It's a great idea to have a 'series...' and I look forward to more.

Sep 11, 2010. 11:11 PM [REPLY](#)

I used to use the 555 and 556 way back in the early 80s and this 'ible has got me going to revisit those days with a few 555 projects.

I wouldn't mind reading a quality audio amp project with minimal components.



SW says:
Seems like you're missing at least a connection in your diagram. Without a connection to V+ to charge up the capacitor, your circuit will be permanently stuck in the "On" position.

Sep 9, 2010. 3:54 PM [REPLY](#)

Also the choice of the 100k resistor at pin 7 means the Off time will be about 10 times the On-time. Perhaps a bit more about calculating exact component values to get precise frequencies will be helpful!



rabbitkillrun says:
The diagram doesn't match up with your breadboard, or at least, the bit around the 12K Ohm resistor is a bit confusing. Also, you say 10K in the part list but 12K from then on.

Sep 10, 2010. 8:35 AM [REPLY](#)



Irwind Kee says:
my first project use this chip at school long time ago

Sep 10, 2010. 7:03 AM [REPLY](#)



Patman27 says:
Thank you so much for starting this, I was just thinking today about learning about an IC I yanked from an old UPC. Made me think of this. :)

Sep 9, 2010. 6:09 PM [REPLY](#)



Saturn V says:
So it's like the red pill?

Sep 9, 2010. 5:17 PM [REPLY](#)



WerdnaN says:
Maybe a guide on the LM386 amp?

Sep 9, 2010. 3:44 PM [REPLY](#)



1up says:
I liked the tutorial; we need more of those. ;) I think you should definitely cover the op-amp. It's a pretty useful little guy.

Sep 9, 2010. 2:15 PM [REPLY](#)



SPGWhistler says:
or an H-Bridge

Sep 9, 2010. 12:37 PM [REPLY](#)



Electronics Blurred says:

How about giving us a guide on 4017 (Decade counter) & 741 (Op-Amp)
;)

Sep 9, 2010. 11:25 AM **REPLY**